Shape memory alloy (Bending Wire)

1. Introduction

This project evaluates control methodologies for an SMA wire actuator modeled via Lagoudas thermodynamics under constant bending. The performance of a conventional linear PI loop is analyzed, highlighting severe tracking lags caused by non-linear phase transformation hysteresis. A feedforward control strategy utilizing First-Order Reversal Curves (FORC) is then introduced to solve the delay problem and ensure precise trajectory tracking.

Austenite Young's modulus (Ea)	70 GPa
Martensite Young's modulus(Em)	30 GPa
Maximum Transformation Strain (et)	.045
Austenite Starting Temperature (As)	280 K
Austenite finishing temperature (Af)	300 K
Martensite Starting Temperature (Ms)	275 K
Martensite Finishing Temperature (Mf)	256 K
Slope of Austenite Limit Curve (Ca)	8 MPa/K
Slope of Martensite Limit Curve (Cm)	8 MPa/K
Heat Capacity At Constant Pressure(Ka)	400 J/Kg.K
Density (Rho)	6500 Kg/m^3
Diameter of The Wire (D)	.5 mm
Length of The Wire(L)	100 mm

Table (1) The wire parameters

2. The hysteresis of the shape memory alloys

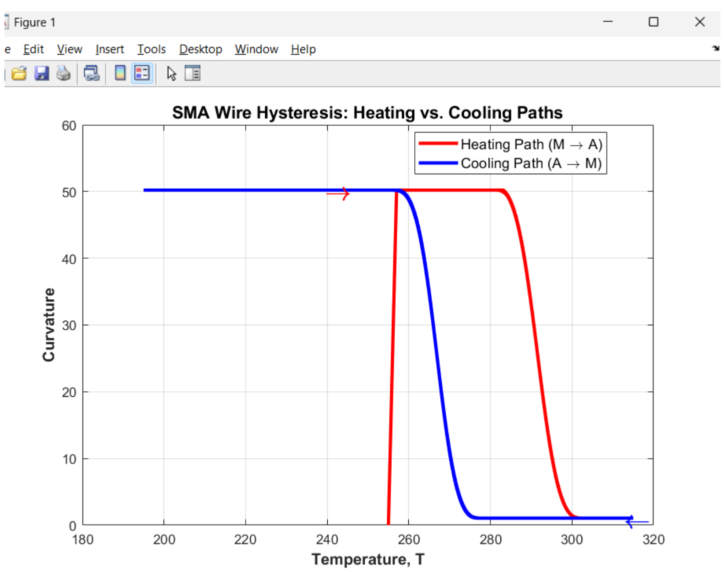

Figure (1) The hysteresis loop of the SMA wire

Shape memory alloys have many benefits, but from a control perspective, their main drawback is hysteresis. To demonstrate how this hysteresis works, a sinusoidal temperature input was applied to the model under a constant load, using an amplitude of 70 K and a bias of 255 K. At the start of the simulation, the bending load applied to the tip of the wire causes an instant jump in curvature to 52 m^{-1}. As the temperature rises from 255 K to nearly 280 K—which is the Austenite Start (As) temperature—the wire shape does not change at all. Once it passes 280 K, the wire contracts and recovers its original shape, pulling the load. During the cooling cycle, as the temperature drops from 320 K down to 275 K, the wire behavior stays completely flat. From 275 K down to 256 K—the Martensite Finish (Mf) zone—the wire loses its stiffness and yields to the bending load again. This complete cycle demonstrates the true hysteresis of the system, showing that the wire never reacts instantly to temperature changes.

3. The linear negative gain PI position control

As shown previously, the wire was bent to $52m^{-1}$., establishing a controllable curvature range from approximately 1 m^{-1} to 52 m^{-1}. This section demonstrates how a conventional linear PI controller performs when applied to this highly non-linear system, building on the hysteresis characteristics observed in the previous section.

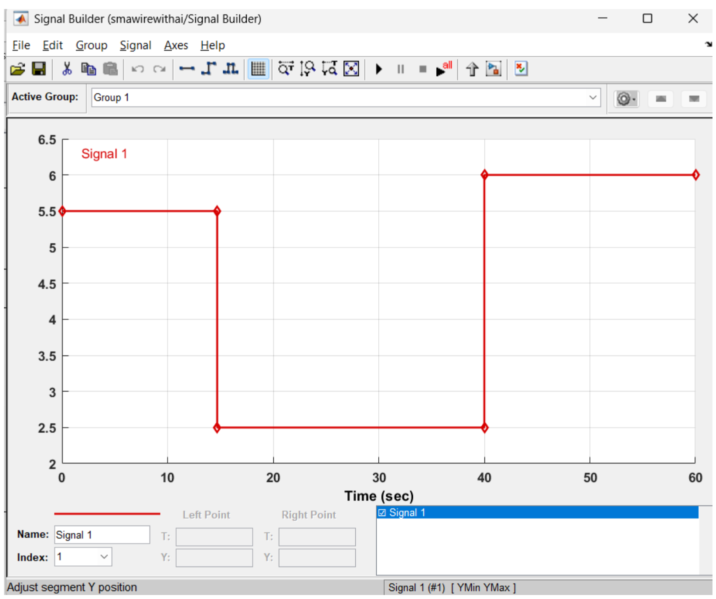

Figure (2) the input signal of the system

The reference input signal shown in Figure 2 starts at 5.5 $m^{-1}$for 15 seconds, drops to 2.5 $m^{-1}$for 25 seconds, and finally increases to 6 m-1 for the remainder of the simulation. A negative gain position controller is implemented because the wire contracts as the temperature rises and yields to the mechanical load as the temperature cools down.

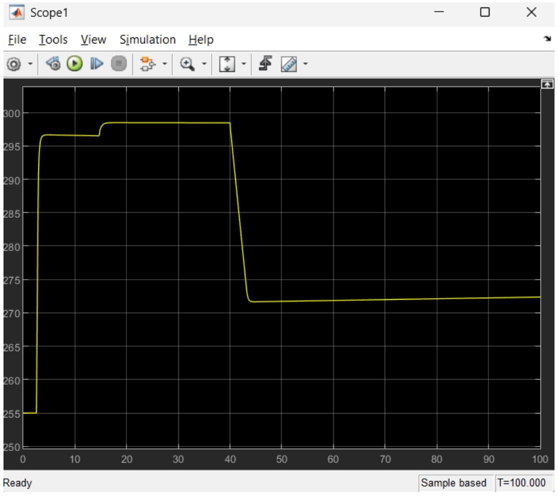

Figure (3) The input temperature plot

The tracking error analysis highlights the severe limitations of using a linear controller on a hysteretic system. During the heating phases, the PI controller experiences an initial delay because it must wind up the temperature past the Austenite Start threshold before any physical curvature change occurs. The performance degrades significantly during the cooling phases, where a pronounced tracking delay of several seconds is observed. This lag occurs because the linear controller cannot inherently anticipate the multi-valued nature of the hysteresis loop, causing the calculated control effort to fall behind the required thermal trajectory needed for shape recovery. Crucially, this simulation assumes an ideal temperature source without an actual heating system, meaning this delay represents only the mathematical error calculation before commanding a physical actuator. Consequently, in a real application, the actual tracking delay will be substantially higher depending on the thermal mass and time constants of the electro-thermal system.

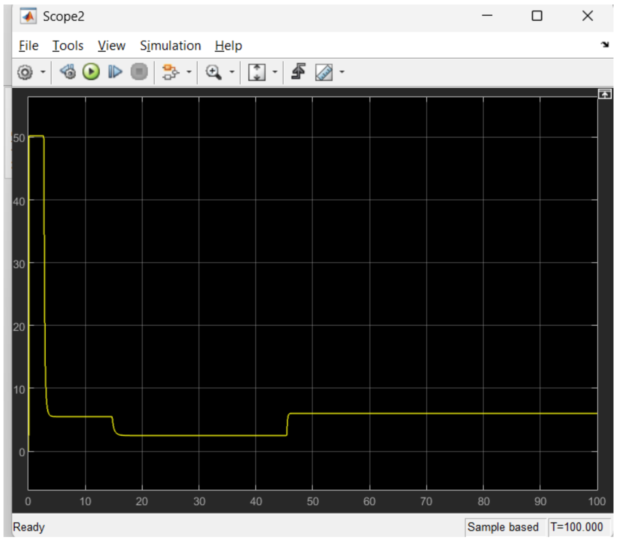

Figure (4) The output curvature plot

The temperature plot confirms significant tracking delays during both the first and final reference steps. This persistent delay issue cannot be resolved simply by tuning the PI controller gains. Attempting to eliminate the lag via aggressive feedback tuning ignores real-world physical constraints, such as the maximum rate of heat transfer within the wire. Ultimately, these tracking limitations are fundamentally rooted in the inherent hysteresis of the material itself.

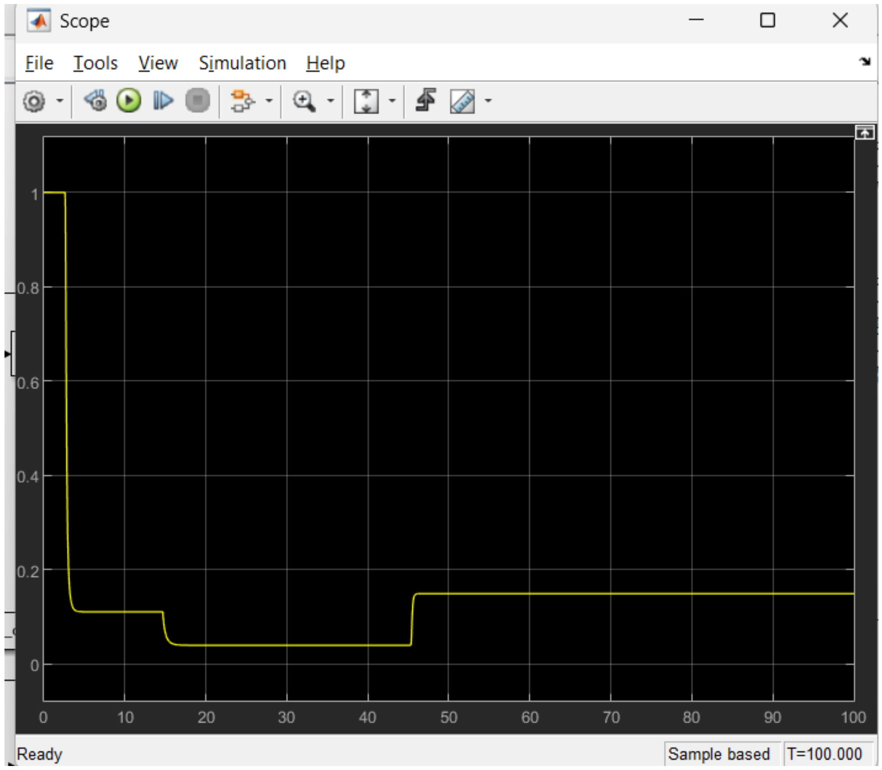

Figure (5) The martensite phase fraction plot

4. The FORC based feedforward control

To overcome the tracking delays caused by material hysteresis and the limitations of linear PID control, a feedforward control strategy was implemented. This controller provides the exact temperature required to follow the desired trajectory. Because the system switches between distinct heating and cooling paths, a core challenge is ensuring the algorithm does not confuse the two different temperature values that exist for a single output curvature (for example, when moving from a maximum bending position of 52 m^{-1}down to 10 m^{-1}, and then heating back up to 15 m^{-1}). To achieve this, the system must be mapped using First Order Reversal Curves (FORC). The wire is heated to a fully austenitic state, cooled to a specific reversal temperature, and then reheated back to full austenite. This process is repeated across gradually decreasing cooling temperatures to comprehensively capture the material physics. For this model, 50 distinct simulation samples were taken across a thermal range from 320 K to 255 K to generate the complete FORC data set.

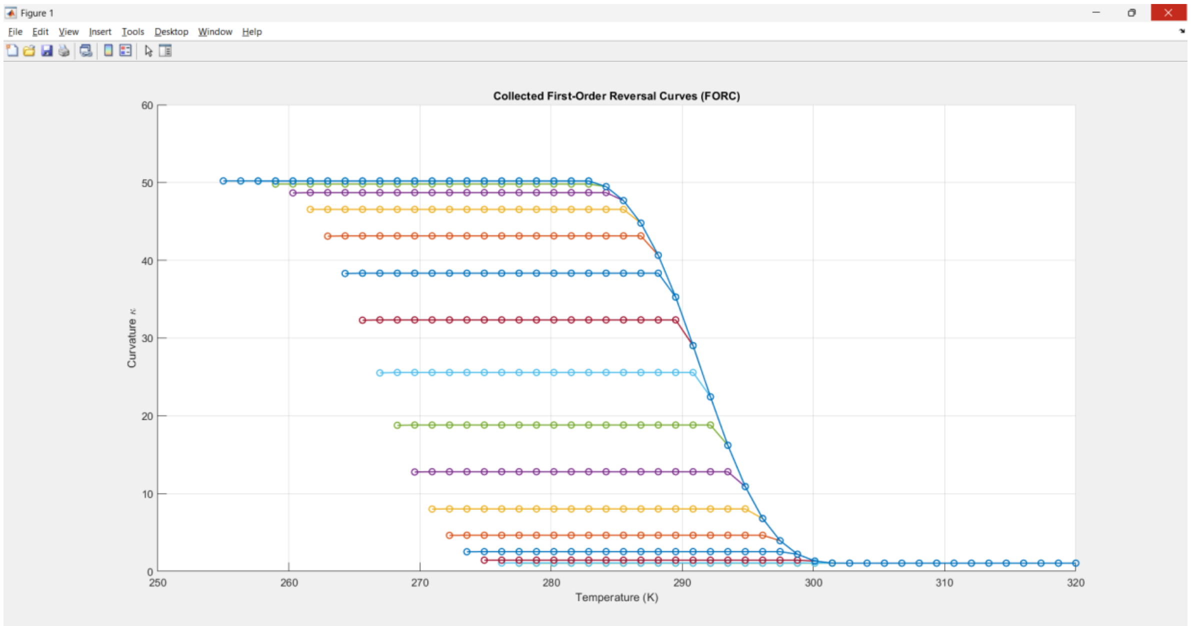

Figure (6) FORC measurement method

This plot captures the system physics at a load the bends the wire to 52 m^{-1}curvature from this data set we are extracting the cooling and heating paths.

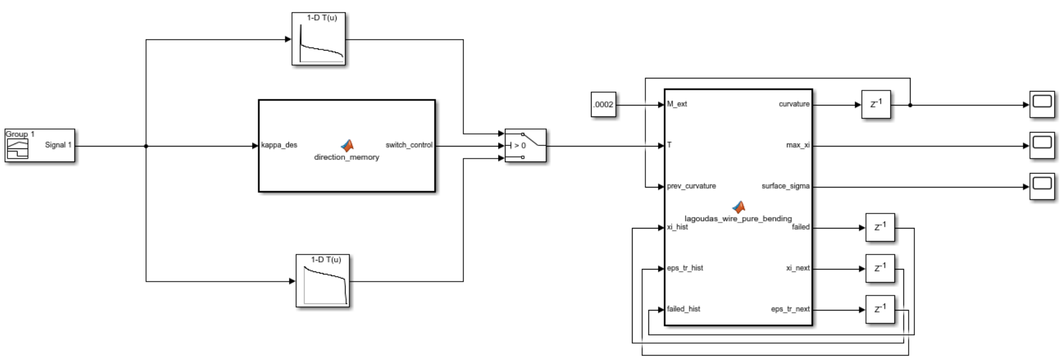

Figure (7) The Feed forward controller

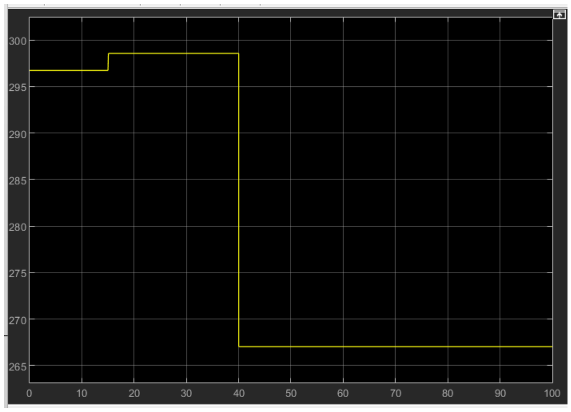

Figure (8) the input temperature to the system.

As shown, the feedforward system targets the predefined temperature at which the desired curvature is achieved without delay. While these dynamics will change in a practical application utilizing Joule heating—since it is physically impossible to jump to a target temperature instantly—the feedforward approach will still be faster than a PID controller. This speed advantage exists because the feedforward system immediately knows the exact target temperature required for a specific curvature, rather than relying solely on the lagging error calculations of a feedback loop.

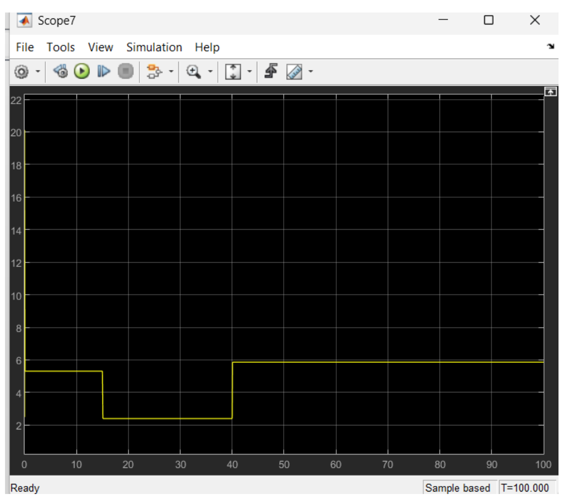

Figure (9) The output curvature

The results illustrate the ideal tracking performance of the feedforward controller; however, this instantaneous response is physically unrealistic. In a real-time experimental setup, or even within a simulation that models Joule heating, the curvature cannot change instantly. This limitation occurs because thermal mass, electrical resistance, and convective cooling rates prevent immediate temperature jumps, meaning that actual system response times will always be governed by thermal transition delays.

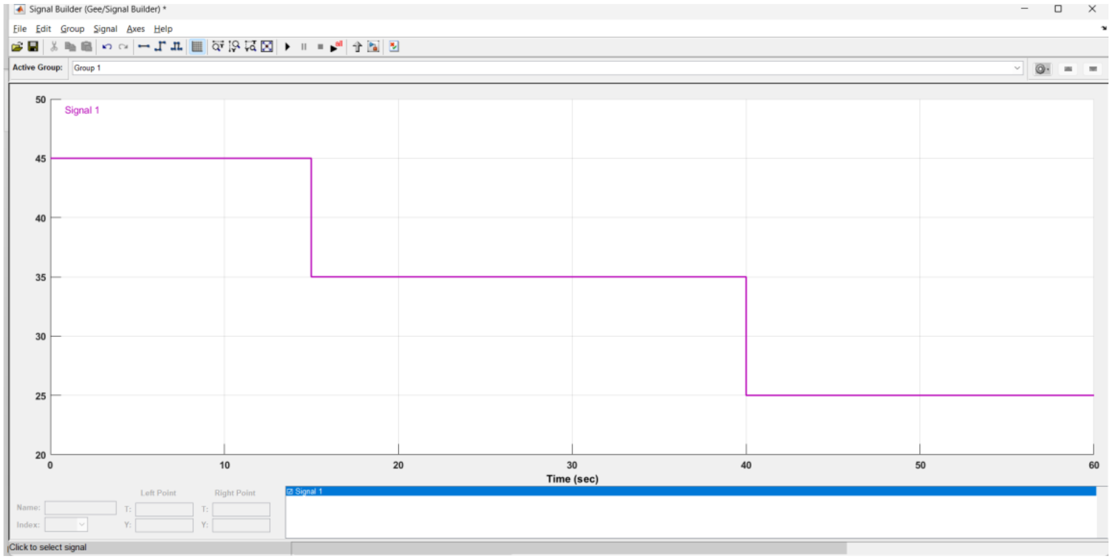

Figure (10) testing with another trajectory

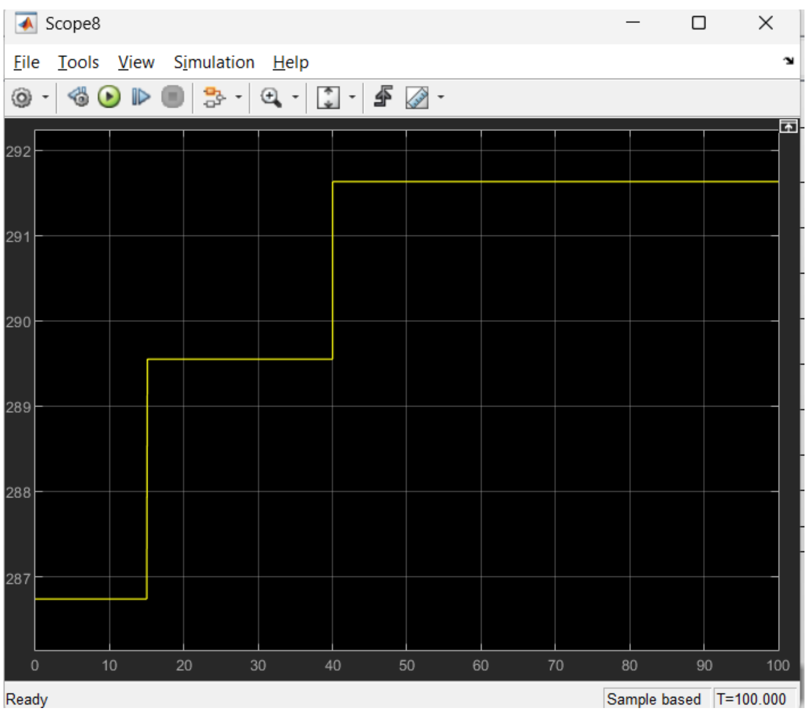

Figure (11) The input temperature plot of the new trajectory

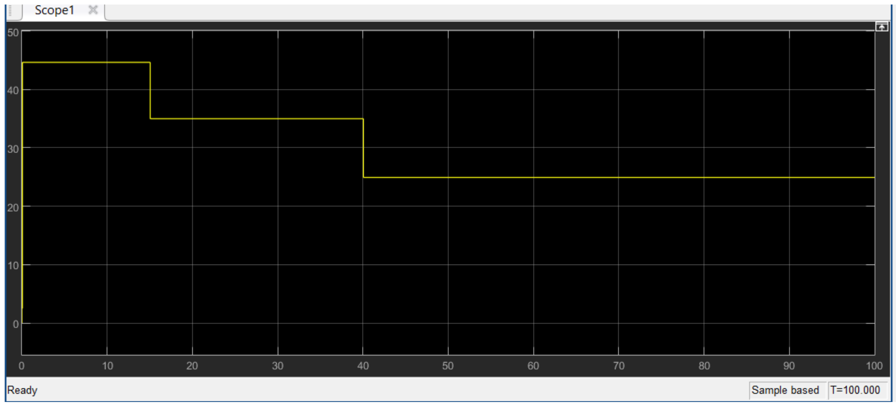

Figure (12) The output curvature of the new trajectory

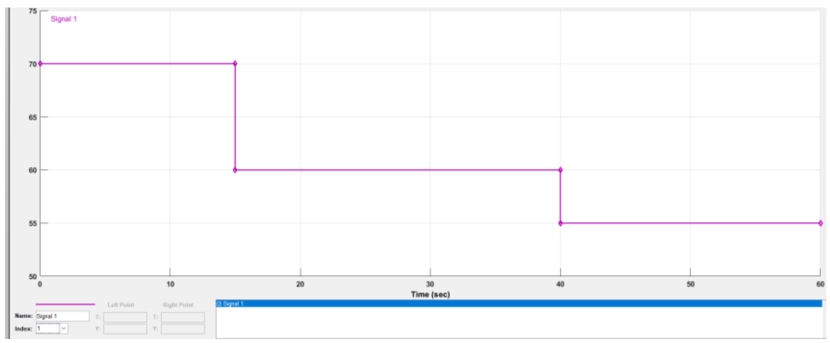

Figure (13) Another out of bounds trajectory signal

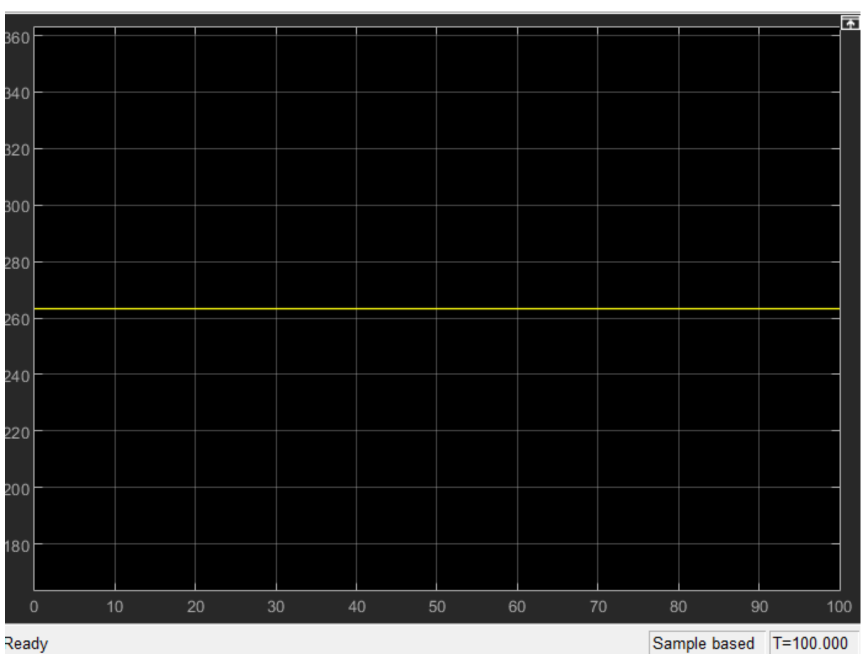

Figure (14) The unchanging temperature plot of the out of bound trajectory

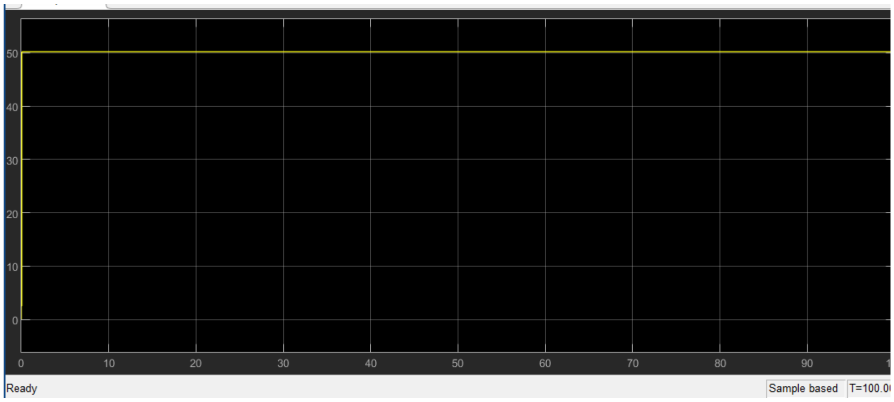

Figure (15) The wire hits its limit it can’t be bent more

5. The shortcomings of the model

This documentation acknowledges two primary limitations of the current idealized model. First, a physical Joule heating system was not modeled to supply the thermal energy required for actuation; instead, temperature was introduced as a direct input signal, neglecting the fundamental thermodynamic laws of heating and cooling. Second, the First-Order Reversal Curve (FORC) data was extracted under a single, constant mechanical load, meaning the current lookup table configuration cannot adapt to varying force conditions. The subsequent two documents will address these limitations to improve the model's accuracy and robustness.

6. The conclusion

In conclusion, this document evaluated the impact of shape memory alloy (SMA) hysteresis on precise position control. The conventional PID controller demonstrated significant tracking limitations and delays due to its inability to anticipate the material's non-linear phase transitions. In contrast, the First-Order Reversal Curve (FORC) lookup table approach proved highly effective as a feedforward controller, delivering fast and accurate trajectory tracking within an idealized system. The next document will be dedicated to integrating Joule heating effects into the simulation to evaluate how the feedforward controller performs under realistic electro-thermal and heat transfer dynamics.